Lets build out the conceptual functions of the Roombox, with every single thing/funcion/capability it needs to cover - whether it is a thing that sits inside the Roombox physical space, or a thing that is placed in the room somewhere inline with the wall and has an 'arm' to it. (Arm: dedicated path within the room or a bus style path where one 'thicker path' is created going around to string together multiple Kötődoboz points within the room. Here are the ground rules:

Every path in a Room is connected to one central point which is the RoomBox, so it reaches every end-point physically. Every RoomBoy can have a wall mounted touchscreen controller. Additional feater include smartphone style screen finger print verification.

A RoomBox is serviced by the BASE (which is the multi location place from where the main inline is placed, to where the Internet/data connection arrives (basic assumption is that both gets connected in the house close together, with preferred/usual Betáp point located near the Enterance door) and where the BRIDGE Home unit PI5 is installed). These three points are preferably close together, with BRIDGE having a connection to the first point it enters the home area (by routing it physically to the BRIDGE unit or - if location make it unfeasable, the BRIDGENET hardware layer, which is then connected to the BRIDGE unit)

The BASE directly reaches the first (often main) RoomBox location - if other Roomboxes exist, they can either be reached by the BASE directly, or as a bus stop, via the first Roombox and so on, until every room/area is reached and covered - everything else and every every end device from there is reached from a RoomBox.

Note-in rule for BRIDGE: The Three main Layers of the the BASE hardware, (NET, COMPUTER, DATA and MEDIA) can be installed as a whole unit or as separate layers in different location in the house, but Crucially important rule: In case of separated layer installation, the architecture needs a separate path layer that physically connects the layers to be wired to each other exactly as would be if everything got installed as one unit - its important to that the BRIDGE base architecture could be the same regardless of physical layer separation.

An additional point may come into play in the design for the SECURITY layer - that would be for a Third Party Security Provider to install its Central device placement (if users want a third party professional security service in the future) Note that these kind of installations are often taking place farther away from the Entrance by design, and require a star topology reach for the end devices - to make it in line with the home system, the pathway placement for the security system respects the base topology, but follow the RoomBox architecture between rooms with parallel pathlaying to the CHANNEL to reach different rooms in the same manner as everything else - and branch out from the same RoomBox spot towards the endpoints in a room as

everything else, creating another layer of pathway dedicated for this purpose. The main design expects the central unit will be installed relatively closer to the first or second RoomBox, rather than the BASE near the Enterance.

Every BRIDGE COMPUTER layer/unit gets a sizeable touchscreen hardwired into the machine for full access. An optional additional design would include a USB-HID capable thin keyboard that could be under the screen to create a full Terminal control feel with it.

The BRIDGE COMPUTER unit should have a direct pathway with the Elosztószekrény unit besides the Mains power, only for data - This provides the foundation for mains level wired MCB power control or simply mains measuring and datastream for the system about usage, amperage, voltage on different ciruits. This becomes feasble two main ways architectually: either a free row of DIN rail within the Elosztószekrény, or as a separate space under/above the Szekrény (that would still give at least the same inner volume as 1 row of DIN would provide) this space would provide physical access to every circuit, installed just after where every circuit (the gerinc) leaves the szekrény - this provides later additional option to place galvanically separated measuring/smart devices for system wide data feed from individual circuits.

DATA and MEDIA has two outlined versions: the basic pack covers at least half-to-One terrabyte of SSD for consumer media usage and filesharing with it, as a mini media server that can contain every personal files of the home users in a shared home network. This can contain the data but doesnt have the horsepower in it to stream it as (another crucial point:) THE BRIDGE COMPUTER UNIT IS A NOT LOAD BEARING, lightweight COORDINATOR AND SMART HOME SYSTEM CONTROLLER UNIT, that avoids every high computational progresses (streaming, big data transfer, rendering, ect) that could slow down its processes. Therefore, the full pack of this layer features a dedicated mini home server unit with the hardware capacity to push every datastream (4K and such) in the home network so that the coordinator remains lightweight with its operations. User facing data ports for this layer can be installed on the front side of the layer installation (if its placed under the BRIDGE COMP unit as it should have when installed as one unit) but if the installation requires, these ports can be extended a few meters from the layer or from the RoomBox) Additional options include Legacy player hardware (DVD-player, ect)

The BASE and the RoomBoxes reach each other via the CHANNEL, the Spine of the whole architecture, which prefers reaching one of the inner walls in the room, and tries to avoid Channel paths going of the 'outer' walls (a standard room for easier management has two walls on the outer side, perimeter walls of the home that can be shared walls with the neighbours, and two 'inner' wall that have the 'same home' on both sides. Standard preferred layout direction from a RoomBox tries to maintain an L shaped end-point preference with its Arms, reaching preferably two walls and three corners (on one is where the RoomBox is installed, and a wall next to it, which is hopefully a shared wall with another room in the Home and logically the next RoomBox that can be placed in that room to folow the pattern.)

Theoretically the first RoomBox is the place where every function can appear - either to service and power the Roombox itself that distributes its function within the room from that point, or as passing wires for the next RoomBox. I tried to collect every distinct, separately achievable theoretical functions that the RoomBox could provide (review every BRIDGE and OmniComm document to check if anything is missing):

- connecting main circuit distribution point for main network, so it distributes towards 230V Sockets, Switches, Lamps (every lamp separately reached with wired connection directly reaching the RoomBox)
- connecting other 230VAC based point like Redőny, Háztartási gépek (if it makes sense in the room) or any dedicated mains powered device the room has to accomodate
- Managing transformation to differnet power levels (to mainly 24/12/5VDC from the 230V mains
- Managing transformed power level wires distribution to end point
- relays, controller units, ect)
- Contains LV operated relays for controlling (switch/dimming) mains lamps
- Distributes LV operated ambient/mood lights
- functions as distirbution switch point roomwide for KNX control (and the other one we already included)
- Acts as the third party security room wide distribution point
- Pathways reaching every door and window besides the SECURITY layer
- Poweres an ESP or PI based sub controller unit (sub to BASE) that is wired to the BRIDGE computer unit so that the computer can control the RoomBox via wired to RoomBox controller unit that monitors and capable to set relay based automations in the room
- has a servicable wired line to media systems via HDMI and (appropriate Audio cables to create the foundations for a home or room wide sound system - we havent worked on it specifically, ideas are welcome)
- Has usb data based capabilities to ditribute USB-A 3.0 or Type-C based wired lines for easy data access
- Has hardware (a Ethernet PoE switch usually in function) to receive UTP cables for further distribution within the room or to send it toward the next room
- Has Wifi AP/Mesh capabilties with one or multiple nodes per room to create dual wifi coverage (5 and 2.4 GHz)
- Has room wide Bluetooth capability with one or multiple nodes per room
- (multiple nodes will have emergent abilities if applied correctly)
- Has Thread order Router and Wifi hardware options dedicated for Matter communication
- Has separate hardware for Zigbee systems
- Can act as a 2.4 GHz Gateway for USB-HID devices

- has 434 and 868 MHz transceivers, and other lower priority communicational hardware for secondary comm types
- Note: LoRa capabilites are only provided for the BASE unit)
- Has pathways to specially created end point locations in the room for different sensors that can feed raw data to the system, collected room-by-room
- Has UTP PoE cable and LV power based pathways to end points in the room for smart- and other cameras
- Can have access to the gardening/watering systems (in practice this belongs to an outdoor RoomBox installed
- Has pathways reaching the thermostat and the (Fűtésrendszer)
- Has pathways reaching the doors or other custom places to provide touch screen based controlling panel capabilities
- Has pathways to reach end points designed for special/custom/Health detection sensors (for gases, leakage, smoke that can appear as risk within a room)
- Has internal sensors within the Roombox to monitor the temperature of the boxes
- Emergent capabilites specific end-point after creating the Layers for the wireless nodes - based on the near finalized pathway layers in a Room, an additional extension of the Wifi/BLE capabilities to create multi node based extras like device location detection, RSSI based presense sensing capabilites, small Tag or sticker based, room accurate positioning of items.

TASK: based on these information, create an updated document for the BRIDGE foundational architecture plan creation based on (and incorporating wirtten info/data for the BRIDGE system) (dont use my words, rewirte it into a coherent document)

In that, include an overview between the base parts and elements of the physical system (BASE, with NET, COMP, MEDIA, CHANNEL, ROOMBOX, ENDPOINTS by type and function (and built up layer-by-layer: MAINS, LV, DATA, KNX, SENSOR any other needs to be mentioned) Elosztószekrény, Security Service Center) and with other rules stated (COMP and Elosztószekrény separate pathways for Data and Power, Security pathway reaches Elosztószekrény to bring two dedicated circuits for the SECURITY layer)

If you want. search up our earlier convos regarding everything BRIDGE related with the names of the files you created earlier - if I still have them, I make another Folder within the H: where I dump them all to start reworking every part into a coherent system and into one, all-encompassing BRIDGE System Reference document that provides every info, spec and data created before for it.